

Hawkeye EQMS · Equipment / Calibration · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

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Live backend	https://hawkeye-backend-dev.vercel.app

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1. Module overview

Module	Equipment / Calibration
Pharma purpose	Track every GMP-relevant asset through its calibration lifecycle. Auto-escalate OVERDUE items. Failed calibrations trigger QUARANTINED status so the asset is blocked from production use.
Compliance basis	ISO 9001:2015 §7.1.5 (monitoring & measuring resources) · 21 CFR 211.68(b) (equipment qualification) · ICH Q7 §5.3
Hawkeye routes	/asset-management (frontend) · /api/equipment (backend)
Hawkeye model file	backend/src/models/EquipmentModel.js
Tenant module flag	modules.ASSET_MANAGEMENT

Equipment moves ACTIVE ↔ UNDER_CALIBRATION ↔ INACTIVE / OUT_OF_SERVICE / QUARANTINED → RETIRED. Each calibration event captures performedBy + performedAt + result (PASS/CONDITIONAL/FAIL) + certificateRef + nextDueDays. FAIL → calibrationStatus=OVERDUE + status=QUARANTINED automatically.

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. MET = parity, PARTIAL = ships but with caveats, GAP = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
Auto-numbered equipment (EQ-YYYY-NNNN) GMP traceability	equipmentNumber pre-save generator.	MET	

Pharma expectation	Hawkeye implementation	Status	Note
Equipment types (ANALYTICAL_INSTRUMENT / PRODUCTION_EQUIPMENT / UTILITY / MEASURING_DEVICE / IT_SYSTEM / OTHER) —	equipmentType enum.	MET	
Calibration frequency in days + next-due tracking 21 CFR 211.68(b)	calibrationFrequencyDays (default 365) + nextCalibrationDue computed from last calibration.	MET	
Calibration history (all events) 21 CFR 211.68(b)	calibrationHistory[] subdocument with date / performedBy / result / certificateRef.	MET	
Status enum: ACTIVE / INACTIVE / UNDER_CALIBRATION / OUT_OF_SERVICE / RETIRED / QUARANTINED —	All 6 supported.	MET	
calibrationStatus enum: CURRENT / DUE_SOON / OVERDUE / NOT_REQUIRED —	All 4 supported. Auto-derived from result + nextDue.	MET	
Failed calibration auto-quarantines + blocks production use 21 CFR 211.68(b)	Calibration result=FAIL → calibrationStatus=OVERDUE + status=QUARANTINED.	MET	
OVERDUE alert when calibration past due ICH Q7 §5.3	calibrationStatus=OVERDUE auto-set; UI badge shows. No notification scheduler today.	PARTIAL	Add cron + notificationOutbox
Soft-retire (RETIRED) with decommissionedAt —	DELETE /:id sets status=RETIRED + decommissionedAt; record retained.	MET	
Linked to batch records (equipmentUsed[]) 21 CFR 211.188(b)(7)	Cross-referenced via batch- records.equipmentUsed[].equipmentId.	MET	

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026`. Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()`.

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Lars Nilsson Maintenance Engineer (admin)	<code>maintenance@novex-pharma.demo</code>	Adds equipment, performs + records calibrations, transitions status, retires assets.	ACTIVE UNDER_CALIBRATION OUT_OF_SERVICE RETIRED
James Thompson Head of QA (admin · approver)	<code>qa.head@novex-pharma.demo</code>	Reviews QUARANTINED assets, signs return-to-service after re-calibration.	QUARANTINED ACTIVE (return)
Aisha Patel QC Lab Lead (admin)	<code>qc.lab@novex-pharma.demo</code>	Owns analytical instruments; performs calibrations on HPLCs, balances.	ACTIVE UNDER_CALIBRATION

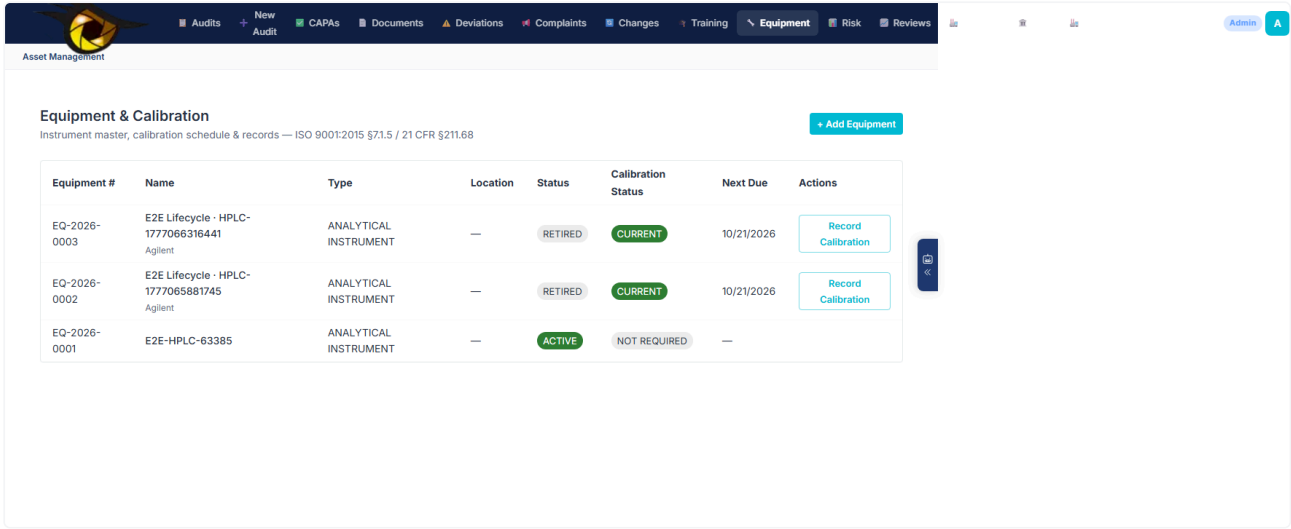
4. Feature catalogue (click-by-click)

4.1 Equipment register

What it does	Lists every asset with type / status / calibrationStatus / nextCalibrationDue.
Menu / route	/asset-management
Available to roles	any tenant viewer
Backend API	GET /api/equipment

CLICK-BY-CLICK

- NAVIGATE** Click 'Equipment' in the top nav
→ Page renders
- WAIT** Spinner clears
→ Rows visible



state-screens/equipment-list.png

4.2 + Add Equipment dialog

What it does	Capture a new asset.
Menu / route	/asset-management · top-right '+ Add Equipment'
Available to roles	maintenance · admin
Backend API	POST /api/equipment

CLICK-BY-CLICK

- CLICK** Click '+ Add Equipment'
→ Dialog opens
- TYPE** Fill name
→ required
- CLICK** Pick equipmentType
→ 6 enums
- TYPE** Fill manufacturer + model + serialNumber + assetTag
- CLICK** Tick requiresCalibration (default true)
→ drives nextCalibrationDue
- TYPE** Fill calibrationFrequencyDays (default 365)
- CLICK** Pick owner (user dropdown)
→ Objectid
- CLICK** Click 'Save'
→ Row appears with status=ACTIVE; equipmentNumber=EQ-YYYY-NNNN

FIELDS CAPTURED

Field	Required	Validation / values	Notes
name	Yes	string	
equipmentType	No	ANALYTICAL_INSTRUMENT PRODUCTION_EQUIPMENT UTILITY MEASURING_DEVICE IT_SYSTEM OTHER	default MEASURING_DEVICE
model	No	string	
manufacturer	No	string	
requiresCalibration	No	boolean	default true
calibrationFrequencyDays	No	number	default 365
ownerId	No	ObjectId	

4.3 Begin calibration (ACTIVE → UNDER_CALIBRATION)

What it does	Mark asset as under calibration.
Menu / route	Equipment row · 'Begin Calibration' button
Available to roles	maintenance
Backend API	PUT /api/equipment/:id (status=UNDER_CALIBRATION)

CLICK-BY-CLICK

1. **CLICK** Click 'Begin Calibration'
→ Status flips

4.4 Record calibration

What it does	Capture a calibration event with result + certificate.
Menu / route	Equipment row · 'Record Calibration' button
Available to roles	maintenance
Backend API	POST /api/equipment/:id/calibration

CLICK-BY-CLICK

1. **CLICK** Click 'Record Calibration'
→ Drawer opens
2. **CLICK** Pick performedAt + performedBy
→ required
3. **CLICK** Pick result (PASS / CONDITIONAL / FAIL)
→ drives next state
4. **TYPE** Fill certificateRef (S3 path or external URL)
→ required
5. **TYPE** (Optional) override nextDueDays (else uses frequency)
6. **CLICK** Click 'Submit'
→ Calibration appended to history; calibrationStatus auto-set; PASS → CURRENT; CONDITIONAL → DUE_SOON; FAIL → OVERDUE + status=QUARANTINED

4.5 Return to service

What it does	Move QUARANTINED / OUT_OF_SERVICE → ACTIVE after a successful re-calibration.
Menu / route	Equipment row · 'Return to Service' button

Available to roles	maintenance + QA approval
Backend API	PUT /api/equipment/:id (status=ACTIVE)

CLICK-BY-CLICK

1. **CLICK** Confirm latest calibration result=PASS
→ QA signs off
2. **CLICK** Click 'Return to Service'
→ Status=ACTIVE

4.6 Retire (soft-delete)

What it does	Decommission an asset. Status=RETIRED + decommissionedAt set.
Menu / route	Equipment row · 'Retire' button
Available to roles	admin · tenant_admin
Backend API	DELETE /api/equipment/:id

CLICK-BY-CLICK

1. **CLICK** Click 'Retire'
→ Confirmation dialog
2. **CLICK** Confirm
→ Status=RETIRED · decommissionedAt set · row remains visible (read-only)

5. Full lifecycle walkthrough — one transaction, end-to-end

One HPLC walked from Add → Calibrate (PASS) → Return to service → Retire.

#	Persona	State transition	Action	API	Observed	Result
1	Lars Maintenance Eng	– → ACTIVE	+ Add Equipment → name='HPLC E2E', type=ANALYTICAL_INSTRUMENT, model=Agilent 1260, calibrationFrequencyDays=180	POST /api/equipment	equipmentNumber=EQ-YYYY-NNNN · status=ACTIVE	PASS

Expected DB state after this step:

```
equipment { _id, equipmentNumber, name, equipmentType: 'ANALYTICAL_INSTRUMENT', model, calibrationFrequencyDays: 180, status: 'ACTIVE', calibrationStatus: 'NOT_REQUIRED' (until first cal) }
```

#	Persona	State transition	Action	API	Observed	Result																																
 Audits New Audit CAPAs Documents Deviations Complaints Changes Training Equipment Risk Reviews Asset Management Equipment & Calibration Instrument master, calibration schedule & records — ISO 9001:2015 §7.1.5 / 21 CFR §211.68 + Add Equipment <table><tr><th>Equipment #</th><th>Name</th><th>Type</th><th>Location</th><th>Status</th><th>Calibration Status</th><th>Next Due</th><th>Actions</th></tr><tr><td>EQ-2026-0003</td><td>E2E Lifecycle · HPLC-1777066316441 Agilent</td><td>ANALYTICAL INSTRUMENT</td><td>—</td><td>RETIRED</td><td>CURRENT</td><td>10/21/2026</td><td>Record Calibration</td></tr><tr><td>EQ-2026-0002</td><td>E2E Lifecycle · HPLC-1777065881745 Agilent</td><td>ANALYTICAL INSTRUMENT</td><td>—</td><td>RETIRED</td><td>CURRENT</td><td>10/21/2026</td><td>Record Calibration</td></tr><tr><td>EQ-2026-0001</td><td>E2E-HPLC-63385</td><td>ANALYTICAL INSTRUMENT</td><td>—</td><td>ACTIVE</td><td>NOT REQUIRED</td><td>—</td><td></td></tr></table> state-screens/equipment-list.png							Equipment #	Name	Type	Location	Status	Calibration Status	Next Due	Actions	EQ-2026-0003	E2E Lifecycle · HPLC-1777066316441 Agilent	ANALYTICAL INSTRUMENT	—	RETIRED	CURRENT	10/21/2026	Record Calibration	EQ-2026-0002	E2E Lifecycle · HPLC-1777065881745 Agilent	ANALYTICAL INSTRUMENT	—	RETIRED	CURRENT	10/21/2026	Record Calibration	EQ-2026-0001	E2E-HPLC-63385	ANALYTICAL INSTRUMENT	—	ACTIVE	NOT REQUIRED	—	
Equipment #	Name	Type	Location	Status	Calibration Status	Next Due	Actions																															
EQ-2026-0003	E2E Lifecycle · HPLC-1777066316441 Agilent	ANALYTICAL INSTRUMENT	—	RETIRED	CURRENT	10/21/2026	Record Calibration																															
EQ-2026-0002	E2E Lifecycle · HPLC-1777065881745 Agilent	ANALYTICAL INSTRUMENT	—	RETIRED	CURRENT	10/21/2026	Record Calibration																															
EQ-2026-0001	E2E-HPLC-63385	ANALYTICAL INSTRUMENT	—	ACTIVE	NOT REQUIRED	—																																
2	Lars Maintenance Eng	ACTIVE → UNDER_CALIBRATION	PUT status=UNDER_CALIBRATION	PUT /api/equipment/:id	Status flips	PASS																																
3	Lars Maintenance Eng	UNDER_CALIBRATION → ACTIVE (CURRENT)	POST calibration with result=PASS, certificateRef='CAL-E2E-001', nextDueDays=180	POST /api/equipment/:id/calibration	calibrationStatus=CURRENT · calibrationHistory[] += entry · nextCalibrationDue=+180d	PASS																																
Expected DB state after this step: equipment.calibrationHistory += { performedAt, performedBy, result: 'PASS', certificateRef, nextDueDays: 180 }; calibrationStatus = 'CURRENT'; lastCalibrationDate; nextCalibrationDue																																						
4	Lars Maintenance Eng	ACTIVE → ACTIVE (returned)	PUT status=ACTIVE (confirm return-to-service)	PUT /api/equipment/:id	Status=ACTIVE (no change)	PASS																																
5	Lars Maintenance Eng	ACTIVE → RETIRED	DELETE /api/equipment/:id (soft- delete)	DELETE /api/equipment/:id	status=RETIRED · decommissionedAt set	PASS																																
Expected DB state after this step: equipment { status: 'RETIRED', decommissionedAt: <now> }																																						

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
(roadmap) Predictive calibration (Wave 3)	ACTIVE UNDER_CALIBRATION	(future)	Equipment detail badge	Predict P(failure) at next calibration from historical trend + usage hours + ambient conditions	Statistical / ML
(roadmap) AI calibration record review	UNDER_CALIBRATION → ACTIVE	(future)	Calibration drawer	Auto-extract values from uploaded certificate (PDF) + flag anomalies	Free Gemini Vision

7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
ACTIVE (CURRENT)	21 CFR 211.68(b) ISO 9001 §7.1.5.2	lastCalibrationDate + nextCalibrationDue + certificateRef + performedBy	equipment + equipment.calibrationHistory
UNDER_CALIBRATION	21 CFR 211.68(b)	Status transition recorded with timestamp	equipment + auditTrail
QUARANTINED (FAIL)	21 CFR 211.68(b) 21 CFR 211.63	Failed calibration record + auto-block from production use	equipment.calibrationHistory + (downstream batch checks)
RETIRED	21 CFR 211.68(b)	decommissionedAt + retention of full calibration history	equipment (soft-deleted)

8. Test results matrix

Suite	Scope	Result	Evidence
eqms-lifecycle.spec.ts · equipment	ACTIVE → UNDER_CALIBRATION → calibration PASS → ACTIVE → RETIRED	PASS	5/5 PASS · eqms-test-results- v2.pdf

9. Known gaps + roadmap

- **OVERDUE / DUE_SOON notification scheduler** — Cron scans nextCalibrationDue; when within 30 days → DUE_SOON notification; when past → OVERDUE + email + Slack. **HIGH**
- **Auto-block from batch records when calibrationStatus != CURRENT** — Batch creation should reject equipmentUsed[] entries pointing to QUARANTINED or OVERDUE assets. **HIGH**
- **Predictive calibration (Wave 3 ML)** — P(failure) at next cal based on history + usage. **MEDIUM**
- **AI cert-extraction on calibration upload** — Free Gemini Vision extracts values from cert PDF. **LOW**